

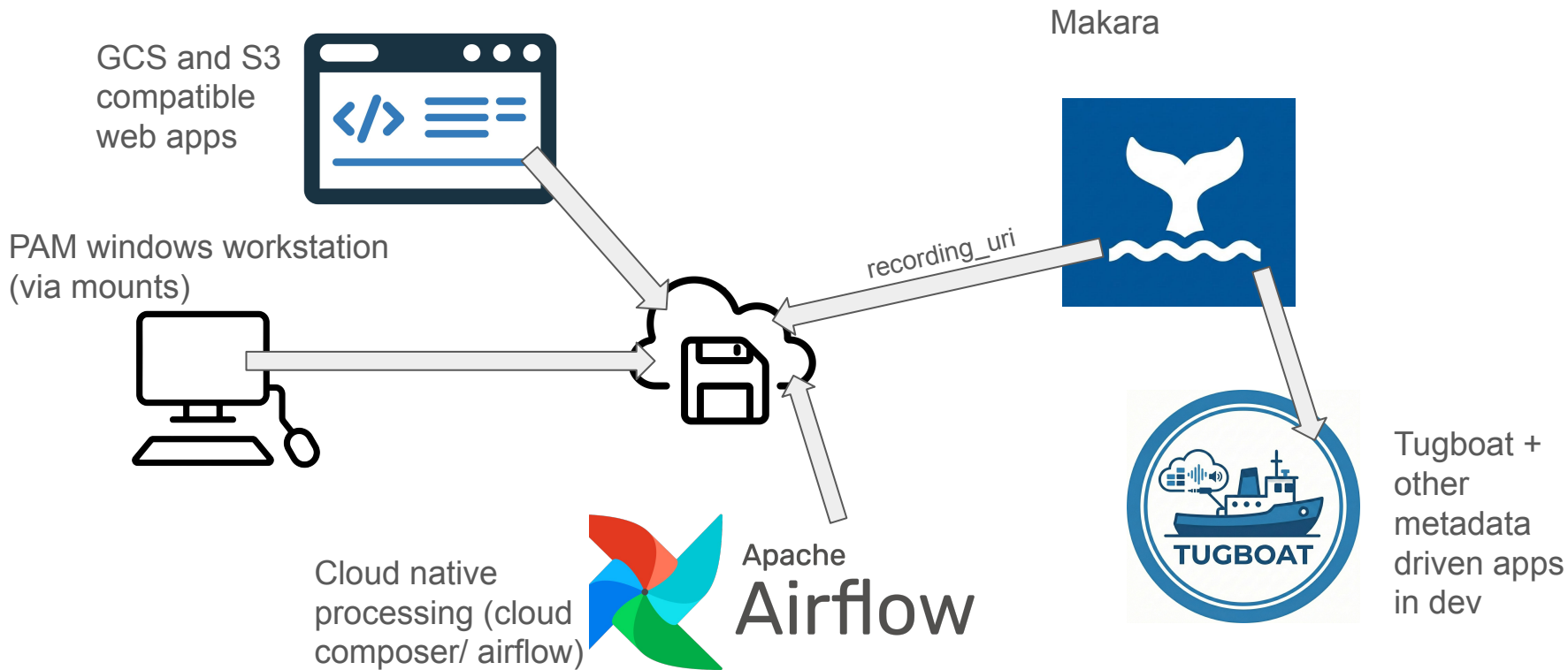
PAM data in cloud storage

Dan Woodrich

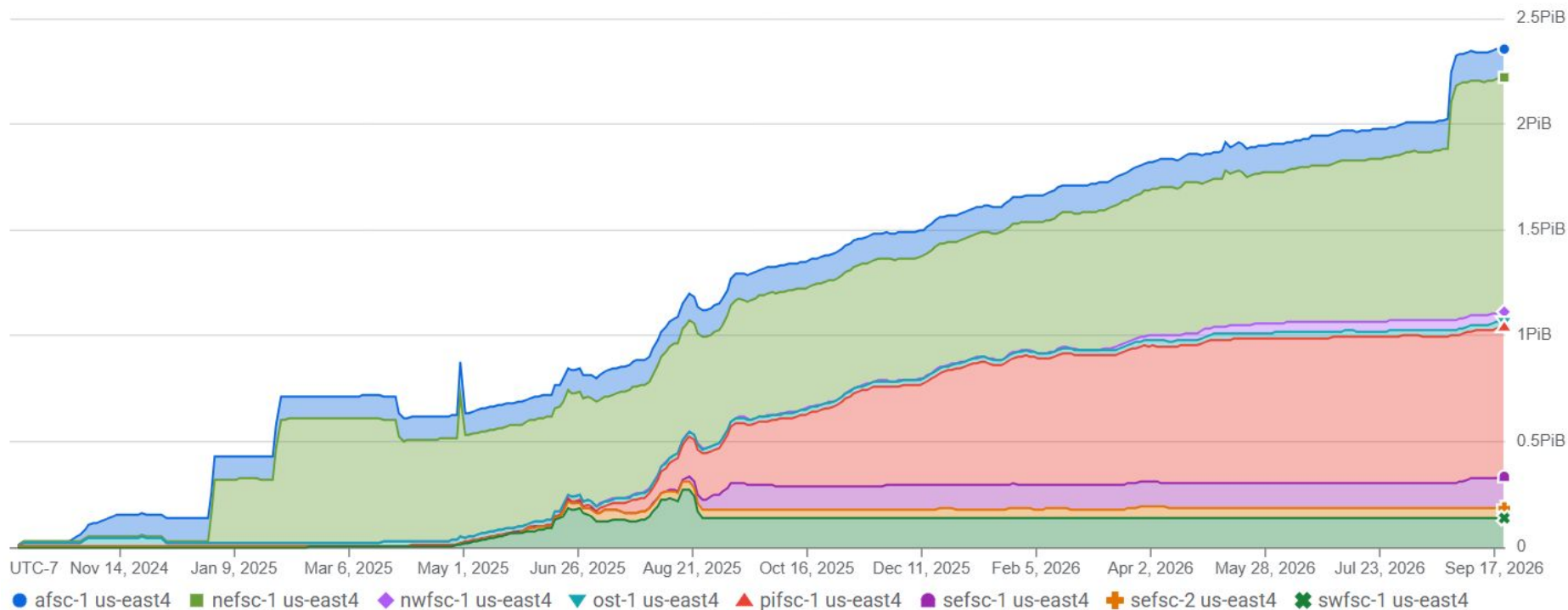
PAM acoustic data

- PAM data are very big: multiple 100TB-PBs data scale PER science center.
- PAM data have high reuse value (active storage)
 - Some regions more than others: NE Atlantic in US has lower mammal biodiversity than Alaska
 - historic data can be 'complete' or in some degree of completion depending on region
- PAM individual data objects are fairly large:
 - On cloud, this has implications for performance

How do you use GCP data



PAM data migration to GCP

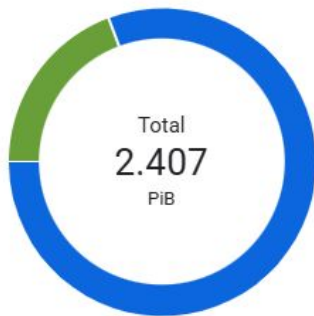


How much data have we uploaded?

Project at a glance

Updated daily ^

Total storage size



● NEARLINE ■ REGIONAL

Total buckets with objects

49

Total objects

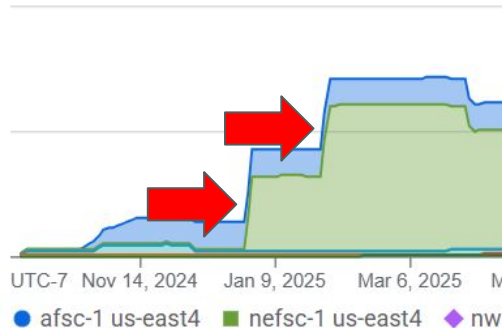
72.748 M

Average object size

35.523 MiB

How to get data to GCP?

- Over the internet:
 - Command line tools: rclone, gcloud storage (formerly gsutil)
 - GUI software
 - Tony Yang [data transfer tool](#)
- Storage transfer appliance
 - Google ships you a hard drive
 - one time bulk transfers



Once data are on GCP, how do you manage it?

- Moving and renaming:
 - Google Cloud Storage (GCP service we use for our data) is atomic, meaning files cannot be modified in place. Renaming or moving a file to a different 'path', is a create/delete operation.
 - Since data are staying in GCP - critical to perform these operations "serverside" - meaning that the data are never downloaded to your computer or VM during this operation.
- Moving and renaming tools that allow you to keep data serverside:
 - CLI: gcloud storage mv, rclone move
 - GUI: GCS move/rename tool:
https://github.com/DanWoodrichNOAA/GCS_Move_Rename_Tool
 - Also TSPI created [a DAG](#) for this
- What not to do!
 - Drag and drop over mounts. Slow, prone to crash. If it must be done, works best on a pam-ww, but in any environment, better to use one of the above listed too;s

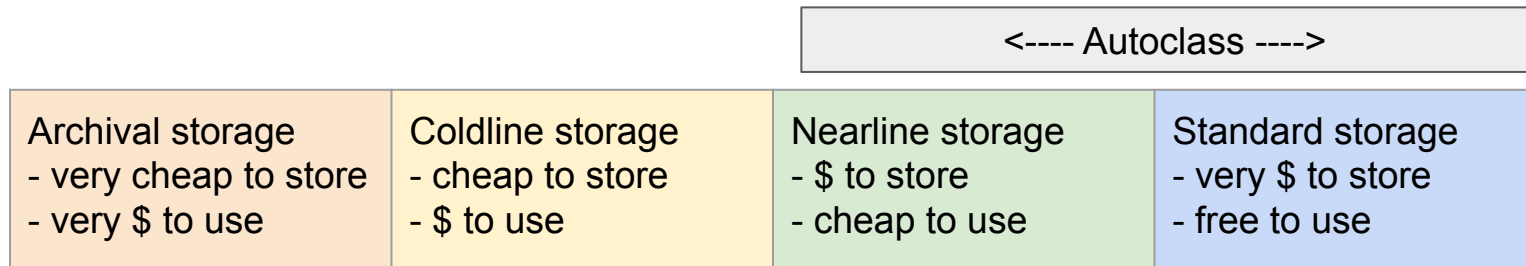
How much are we spending on data in GCP?

- Costs aren't coming down to users, or even science centers at the moment
- NOAA has a persistent discount on cloud costs
- PAM SI admins have control over what storage class to set, in GCP, there are four options along a spectrum, and they control how much the relative cost of use vs storage are for your data.

Archival storage - very cheap to store - very \$ to use	Coldline storage - cheap to store - \$ to use	Nearline storage - \$ to store - cheap to use	Standard storage - very \$ to store - free to use
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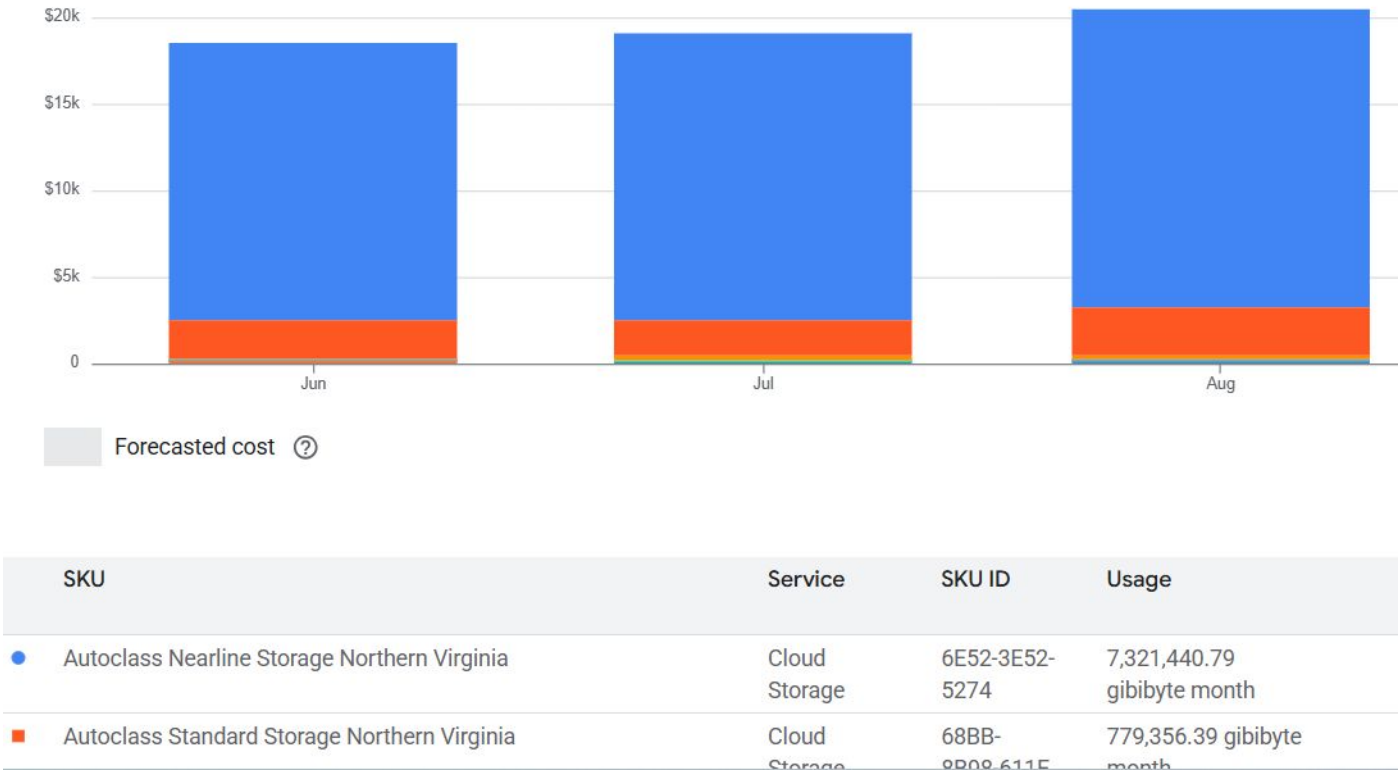
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Spending

- Autoclass
- Our data are mostly in nearline
- Almost all of our spending is related to storage, not use

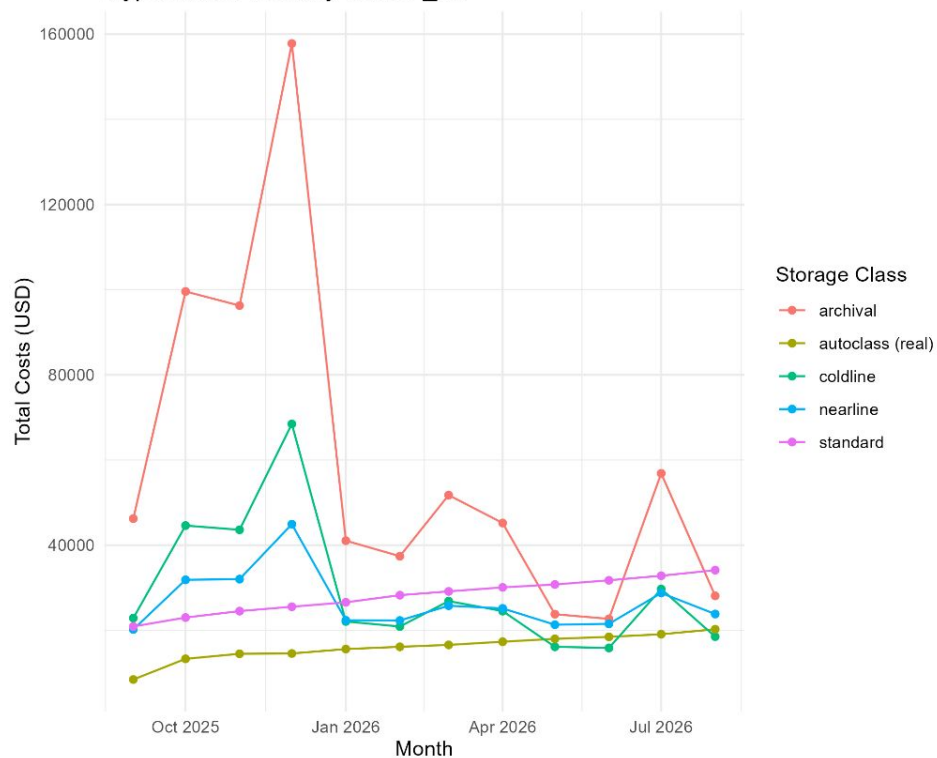


Is autotclass optimal?

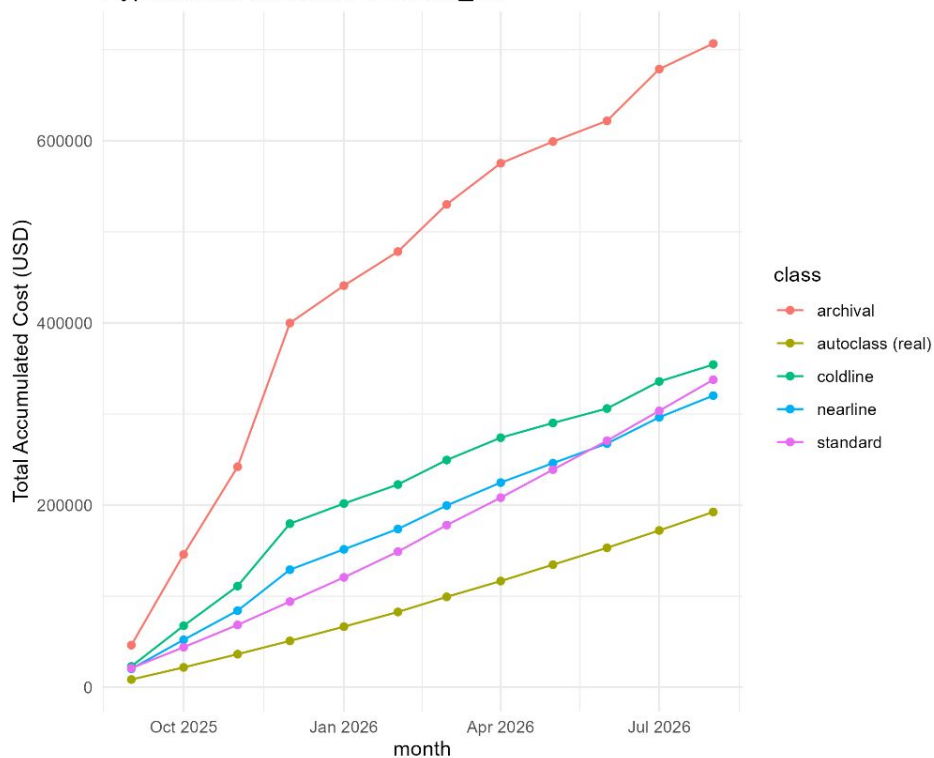
- If we are paying so much to store, should we tier down to colder storage class? nearline across the board or coldline?
- I run an analysis every 6 months to track this.

Is autotclass optimal? Overall: correct

Hypothetical Monthly Costs: _All

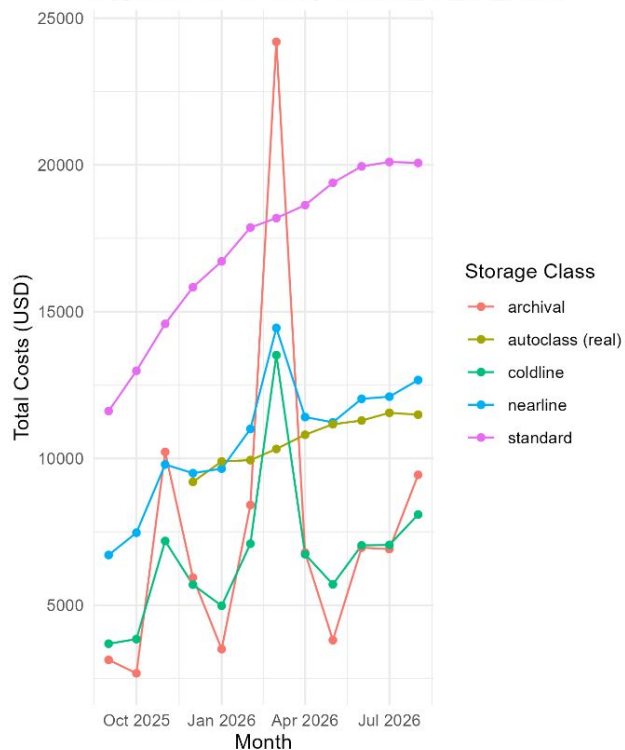


Hypothetical Cumulative Costs: _All

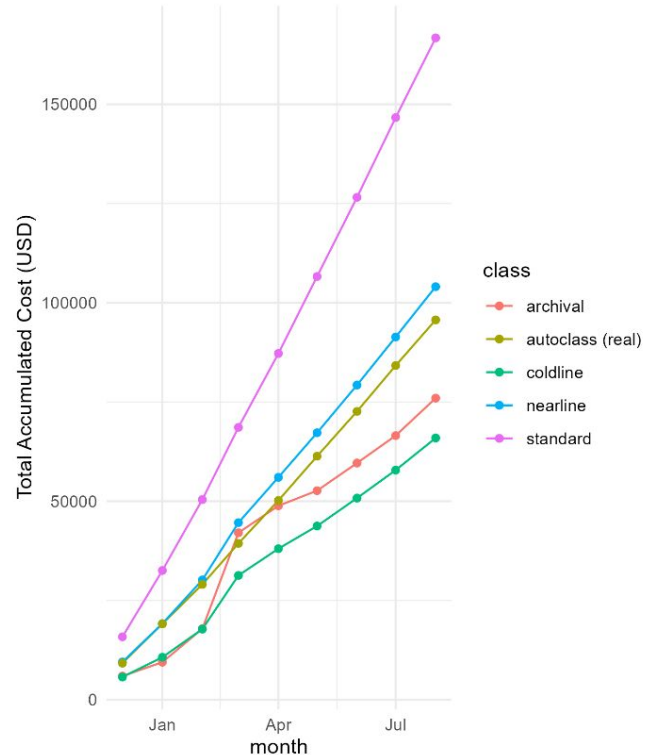


NEFSC is the biggest 'whale' - what about the rest?

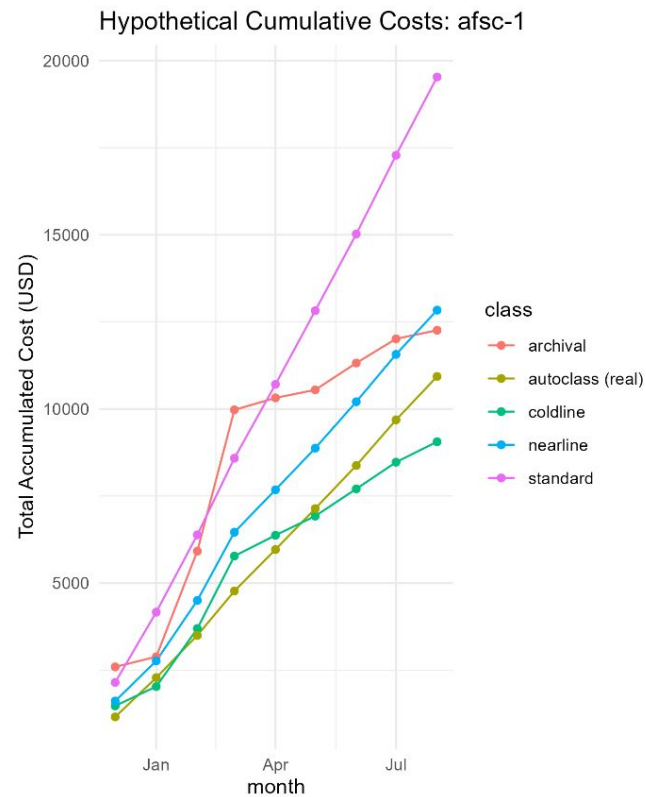
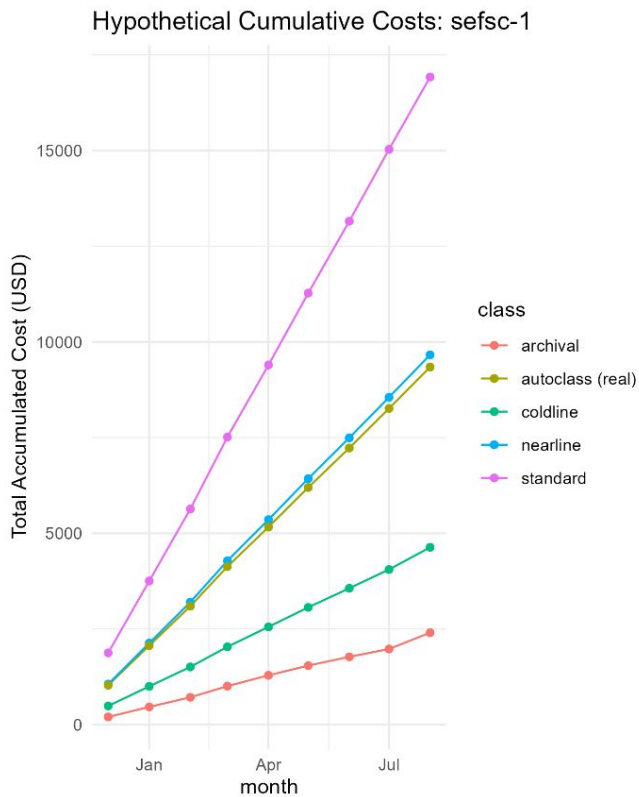
Hypothetical Monthly Costs: _All_no_nefsc



Hypothetical Cumulative Costs: _All_no_nefsc



Individual FMC could have benefited from being colder



Was it the right choice to apply broad autoclass?

- A more granular policy (applying colder classes to colder buckets) up front may have saved money
- Hidden downside of: incentivizes groups to NOT use their data for fear of spend
- Colder classes also come with spending commitments: as use of data ramps up, these are limiting and could lead to high cost spikes, making it harder to rollout new techniques.
- In short - yes!
- Mindfulness will go along way: bundle analyses together while data are 'hot' to save serious \$ for the enterprise.

Cost implications

- OCIO and the PAM leads will need to negotiate how to efficiently support science needs in cloud
- Value of historic data collections - ML training, retrospective analysis, and rerunning older analyses as tools improve: can we ever really be 'done' with a dataset?
- Cloud spending model has natural friction points here
- Admins of our SI project will keep the status quo until told otherwise by leadership!
- Share specific pain points with admin team - helps us deliver easier tools (such as the rename_move tool)

Next steps

- Analyzing GCS to provide structure for more streamlined use with apps and packages
 - See annotation db slide in Perch Hoplite
- Populating Makara is most effective step towards this:
 - recording_uri creates the authoritative link
 - Timezone also critical for structured outputs that are safely comparable and interpretable
- On GCS, make sure files have a timestamp in the name